

WISEMap: Simultaneous Infrastructure and Layout Mapping via Enterprise Wi-Fi

Yichen Wei¹, Yibo Pi^{1†}, Yifei Sun¹, Yihui Li², Dingxiang Peng², Cailian Chen¹

¹Shanghai Jiao Tong University, Shanghai, China

²Ruijie Networks, Fuzhou, China

{ethepherein, yibo.pi, yifei_sun, cailianchen}@sjtu.edu.cn, {liyh, pengdx}@ruijie.com.cn

Abstract

Environment-aware capabilities have been envisioned to enhance Wi-Fi networks in communication, localization, and sensing. However, existing environment-aware approaches often assume prior knowledge of infrastructure (AP orientations and locations) and environment maps, limiting their applicability in real-world deployments. We present WISEMap, a framework that simultaneously reconstructs infrastructure and indoor layout via enterprise Wi-Fi, without prior information. The core challenge lies in the tight coupling between the two tasks: layout reconstruction demands precise infrastructure mapping, which in turn requires accurate mapping of the layout. WISEMap addresses this challenge by alternately optimizing the two tasks and progressively refining their estimates. Real-world evaluations show that WISEMap not only accomplishes both tasks simultaneously but also delivers substantially higher accuracy than the baselines designed for each task, reducing layout reconstruction error by 43–77% and the average AP and user localization errors by 70% and 39%, respectively.

CCS Concepts

• **Networks** → **Wireless local area networks; Wireless access points, base stations and infrastructure; Location based services.**

Keywords

Wi-Fi, Localization, Layout Mapping, SLAM

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1 Introduction

Environment-aware communication is emerging as a promising paradigm for next-generation indoor communication systems. By leveraging environment information, richer channel knowledge can

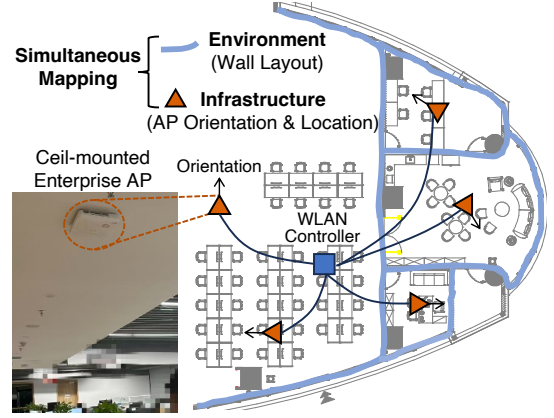


Figure 1: Simultaneous infrastructure and layout mapping via enterprise Wi-Fi, where APs are densely deployed and managed by a WLAN controller.

be extracted to enhance not only communication but also localization and sensing. For instance, a layout map enables the distinction between line-of-sight (LoS) and through-wall paths for more accurate localization [18, 28]. It also supports proactive prediction of channel dynamics for mobile users, allowing adaptive beam steering [27] and seamless handover [30]. Further, indoor environment information significantly improves network management, e.g., coverage planning [24], interference management [22], and network issue debugging [15].

These benefits motivate us to endow existing Wi-Fi networks with environmental awareness. However, most prior works focus on how to exploit environment information and assume that both the physical environment and the infrastructure maps (locations and orientations of APs) are known. In practice, this assumption rarely holds. Network operators often lack accurate knowledge of AP locations, let alone their orientations [3]. Obtaining such information through manual site surveys requires full-area access, which is prone to measurement errors and must be repeated after AP relocation. Moreover, antenna orientations can be difficult to determine when APs are enclosed. Similarly, floor plans may be unavailable, outdated after renovation, or inaccessible when building and network information are managed by different parties. This raises a key question: **can existing enterprise Wi-Fi networks reconstruct both the indoor layout and infrastructure maps via AP-side measurements?** As shown in Fig. 1, the WLAN controller orchestrates APs to overhear regular uplink traffic from all nearby user devices (UEs) and extract channel information from these packets.

[†] Corresponding author.



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Answering this question is challenging due to the tight coupling between the two tasks. Prior RF-based SLAM systems have demonstrated that infrastructure localization and layout mapping can be jointly performed using UWB or mmWave measurements [17, 20]. However, extending this capability to commodity enterprise Wi-Fi is challenging because its substantially noisier angular and ranging measurements make errors in infrastructure estimation and layout reconstruction strongly coupled. Accurate AP location and orientation estimation is a prerequisite for layout reconstruction, and the reconstructed layout, in turn, reveals signal propagation characteristics that refine the estimation of AP location and orientation. Existing Wi-Fi approaches typically address each task separately: they either assume that the other task is perfectly solved [10, 14], or neglect its impact [6, 25, 26]. The former leads to unrealistic assumptions that limit applicability, while the latter sacrifices performance, especially in complex real-world environments.

In this work, we present WISEMap, an alternating optimization framework that simultaneously reconstructs the Wi-Fi infrastructure and the indoor layout using enterprise APs. WISEMap leverages AP-side CSI and RSSI measurements passively collected from regular Wi-Fi traffic. Specifically, WISEMap first estimates AP locations and orientations without a layout prior, producing a coarse initialization. Based on this, it reconstructs an initial layout, which is used to identify through-wall paths and refine localization. This process alternates to progressively reduce errors.

However, realizing this requires addressing several key challenges. First, APs must localize themselves without any prior information. Previous works [6, 26] have explored using inter-AP AoA estimates for self-localization, but our experiments indicate that per-link AoA estimates are highly error-prone in indoor environments. To mitigate random errors, WISEMap leverages a large number of UEs seen by multiple APs as anchors and localizes APs by aligning their respective views on the locations of these common UEs (§2.2). Second, reconstructing wall layout requires estimating paths between AP-UE pairs that reflect off walls. However, random synchronization offsets between APs and UEs prevent accurate absolute range estimation [34]. We observe that the LoS and reflection paths experience the same synchronization error, allowing the range difference between them to remain unaffected. By combining this range difference with the LoS-path length inferred from RSSI, WISEMap can estimate the reflection-path length (§2.3). Third, due to estimation errors in AP and UE locations as well as path properties, the estimated reflection points on walls are highly noisy, making layout reconstruction challenging. WISEMap leverages the expressiveness of signed distance functions (SDFs) to model the complex indoor geometries and recover surfaces that best fit these noisy observations (§2.3). The indoor layout is divided into individual AP regions for reconstruction, which are subsequently fused to form the final layout map (§2.4).

We evaluate WISEMap across four diverse indoor layouts, two of which are operational enterprise networks. Our results show that, compared with the state-of-the-art methods, WISEMap not only enables simultaneous infrastructure mapping and layout reconstruction (as summarized in Table 1), but also achieves higher accuracy on both tasks. Specifically, WISEMap reduces the Hausdorff distance of the reconstructed layout by 43-77%, and, by leveraging

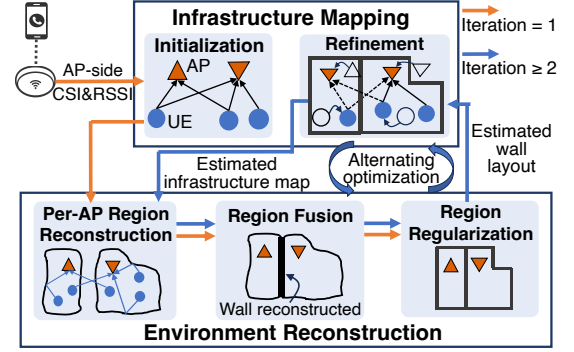


Figure 2: Overview of WISEMap's workflow

the reconstructed layout, further reduces the average AP and UE localization errors by 70% and 39%, respectively.

In summary, we make the following contributions:

- We show, for the first time, that enterprise Wi-Fi networks can simultaneously reconstruct infrastructure and indoor layout maps by overhearing regular user traffic, without any prior information.
- We present WISEMap, an alternating optimization framework that jointly solves infrastructure mapping and layout reconstruction. It leverages the collective power of UE observations to mitigate noise in real-world samples via AP view alignment and SDF-based layout reconstruction.
- We implement and evaluate WISEMap across diverse real-world enterprise deployments, showing significant improvements over the state-of-the-art methods in both reconstruction and localization accuracy.

2 WISEMap Design

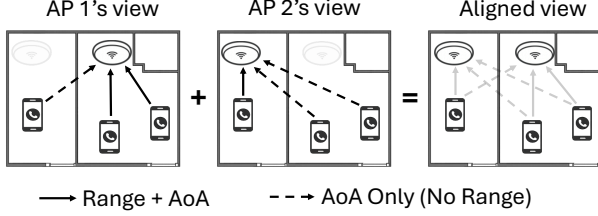
2.1 Overview

WISEMap is an alternating optimization framework that performs simultaneous mapping of Wi-Fi infrastructure and indoor layout. The two tasks are coupled and can be alternately refined: layout mapping enables the distinction of LoS and through-wall paths for better AP localization; conversely, more accurate AP localization can improve the accuracy of layout mapping. Fig. 2 illustrates the workflow of WISEMap that alternately optimizes the two modules below:

- (1) **Infrastructure Mapping:** This module takes CSI and RSSI measurements as input and infers AP locations and orientations by aligning their observations of common UEs across multiple APs. In the first iteration, all paths are considered LoS for localization. From the second iteration onward, once a layout estimate becomes available, WISEMap leverages it to identify through-wall paths and filter out these unreliable paths for ranging.
- (2) **Layout Mapping:** Given the estimated AP locations and orientations, each AP first reconstructs its surrounding region. These per-AP regions are then merged based on their spatial relationships and subsequently stitched together to form a global layout map. We then apply region regularization to refine the

Table 1: Comparison with prior solutions for indoor layout and Wi-Fi infrastructure mapping

Capability	WISEMap	NeRF ² [31]	EchoNeRF [2]	MapFi [26]	UbiLocate [23]	SpotFi [12]
UE localization	✓	✗	✗	✓	✓	✓
AP localization (incl. orientation)	✓	✗	✗	✓	✗	✗
Environment reconstruction	✓	✓	✓	✗	✗	✗
Compatible w/ deployed networks	✓	✗	✗	✗	✗	✗


Figure 3: Infrastructure mapping by aligning AP views based on shared UE observations

reconstruction. The refined layout is fed back into the infrastructure mapping module, and this alternating optimization process iterates until no significant changes occur in either the layout or infrastructure map.

2.2 Infrastructure Mapping

Key Insights: Aligning Common UEs across APs. In indoor environments, each AP has its own local view of nearby UEs, and these local views are naturally linked through common UEs visible to multiple APs. By aligning the observations on these common UEs across APs, we can infer the relative positions between APs and UEs. As shown in Fig. 2, AP 1 and 2 form their respective views of UE locations, and their relative positions can be determined by aligning these individual views.

UE Localization. An Enterprise AP with a planar antenna array can estimate the 3D AoA (azimuth and elevation) of incident signals in its local coordinate system, which defines the AP's orientation. Let the local coordinate system of AP k be spanned by orthonormal basis vectors, $\mathbf{o}_k^x$, $\mathbf{o}_k^y$, and $\mathbf{o}_k^z$, and $\hat{\mathbf{v}}_{l \leftarrow k}$ be the AoA unit direction vector of signal from UE l to AP k in the global coordinate system, where $\mathbf{o}_k^z$ is fixed for flat ceil-mounted APs. Given an estimated distance $\hat{d}$ from AP k to UE l , AP k infers the UE's global location as $\mathbf{p}_{l \leftarrow k}^{\text{UE}} = \mathbf{p}_k^{\text{AP}} + \hat{d}\hat{\mathbf{v}}_{l \leftarrow k}$, where $\mathbf{p}_k^{\text{AP}}$ denotes the location of AP k . To estimate $\hat{d}$, prior work typically relies on time-of-flight (ToF) inferred from subcarrier phase shifts [34], fine timing measurements (FTM) [23], or RSSI-based ranging [8, 12]. We adopt the RSSI-based approach for ranging estimation, as the random timing offsets between the AP and UE in commodity Wi-Fi devices introduce significant errors [34]. While FTM can provide accurate ranging, it requires explicit protocol support and coordination between APs and users. WISEMap extracts channel measurements from regular user traffic without requiring any user coordination. In contrast, RSSI-based ranging via the log-distance path loss model provides a practical solution for single-AP UE localization, where the model parameters are calibrated for the specific deployment environment.

Using single RSSI measurements for range estimation is error-prone. When aggregated across nearby UEs, however, RSSI measurements collectively capture the signal attenuation and average out random errors in individual RSSI measurements. To mitigate short-term fluctuations caused by multipath and dynamic environments, we average each UE's RSSI over a one-second window. We further discard weak signals below a threshold, as they severely degrade both AoA and ranging accuracy.

Aligning Multi-AP Views. After each AP locally estimates the positions of nearby UEs, we jointly infer the AP locations and orientations by aligning their respective views of the common UEs. Specifically, we first optimize the AP locations and orientations to minimize the location discrepancy between common UEs across APs, referred to as *UE location alignment*. In the first iteration, WISEMap has no prior knowledge of the layout and assumes that each AP has a LoS path to all nearby UEs. AP locations and orientations can then be determined by aligning UE locations across APs as follows:

$$(\mathcal{P}1) \min_{\{\mathbf{p}_k^{\text{AP}}, \mathbf{o}_k^x, \mathbf{o}_k^y\}_{k \in \mathcal{A}}} \mathcal{L}_{\text{LoS}} = \sum_{l \in \mathcal{U}} \sum_{k \in \mathcal{A}_l^{\text{LoS}}} \sum_{s \in \mathcal{A}_l^{\text{LoS}}} |\mathbf{p}_{l \leftarrow k}^{\text{UE}} - \mathbf{p}_{l \leftarrow s}^{\text{UE}}|^2, \quad (1)$$

where $\mathcal{U}$ and $\mathcal{A}$ denote the sets of all UEs and APs, respectively, and $\mathcal{A}_l^{\text{LoS}}$ denotes the set of APs in LoS with UE l . It can be shown that $\mathcal{P}1$ is a convex quadratic program, which can be solved optimally.

From the second iteration onward, the reconstructed layout enables us to distinguish direct LoS and through-wall paths. Specifically, a path is classified as through-wall if the line segment between the AP and UE intersects any wall inferred from the previous iteration. While RSSI-based ranging is unreliable for through-wall paths, their associated AoA measurements remain informative. We therefore leverage these paths to perform *UE AoA alignment* across APs. Specifically, we want to minimize the discrepancy between the geometric and measured AoAs for UEs at the through-wall APs, where the geometric AoA corresponds to the incident angle of the through-wall path. To obtain the geometric AoA, we first localize each UE using their direct LoS APs and take the centroid of the resulting estimates as UE position. Using this estimated UE position together with the AP locations, we compute the geometric AoA. The resulting discrepancy between the geometric and measured AoAs of the through-wall paths at these APs can be written as

$$\mathcal{L}_{\text{NLoS}} = \sum_{l \in \mathcal{U}} \sum_{k \in \mathcal{A}_l^{\text{NLoS}}} |\mathbf{q}_{l \leftarrow k} - \hat{\mathbf{v}}_{l \leftarrow k}|^2, \quad (2)$$

where $\mathbf{q}_{l \leftarrow k}$ and $\hat{\mathbf{v}}_{l \leftarrow k}$ are the unit direction vectors of the geometric and measured AoAs at AP k for UE l . For UEs not in LoS with any AP, their measurements are excluded from AP location and orientation estimation. From the second iteration onward, the positions and orientations of APs can then be determined by minimizing the

following combined loss:

$$(\mathcal{P}2) \quad \min_{\{p_k^{\text{AP}}, o_k^x, o_k^y\}_{k \in \mathcal{A}}} \lambda_{\text{LoS}} \mathcal{L}_{\text{LoS}} + \lambda_{\text{NLoS}} \mathcal{L}_{\text{NLoS}}, \quad (3)$$

where λ_{LoS} and λ_{NLoS} are loss normalizers. As $\mathcal{P}2$ is non-convex, to avoid converging to undesirable local minima, we first perform UE location alignment by solving a LoS-only variant of $\mathcal{P}2$ using only direct LoS paths, which generates a globally optimal solution as in $\mathcal{P}1$. We then use this solution as a reliable initialization and refine it with AoA alignment by solving $\mathcal{P}2$ via gradient descent.

2.3 Per-AP Region Reconstruction

We consider the layout as composed of regions covered by individual APs, where each AP independently reconstructs its surrounding area. As shown in Fig. 4, each NLoS reflection path of an AP-UE pair reveals one reflection point, and with a sufficient number of these paths, the resulting points collectively span the environment's surfaces. However, locating a reflection point requires the AP and UE locations as well as the AoA and range estimates of the NLoS path, all of which are subject to estimation errors. Consequently, these reflection points cannot be simply connected to reconstruct the layout. To address this, we leverage the expressiveness of SDFs to model the complex indoor geometries and adapt the model to best fit the estimated reflection points.

AoA and Range Estimation for NLoS Paths. After infrastructure mapping, AP and UE locations and AP orientations are determined. Since reflections significantly attenuate the signal strength, double-bounce paths have much weaker signal strength than single-bounce paths. Therefore, the strongest NLoS path is treated as the single-bounce reflection path. We extend the RAP-MUSIC algorithm [19] to support 3D AoA estimation for NLoS paths. However, we cannot directly estimate the range of a NLoS path because of the random synchronization offsets between the APs and UEs discussed earlier. To address this, we notice that both LoS and NLoS paths experience identical synchronization errors, making their range difference independent of synchronization errors. Thus, we compute the range difference between the LoS and NLoS paths and add it to the LoS path to recover the NLoS path length. Let f_0 and c be the frequency and speed of the signal. As detailed in [34], the phase difference between consecutive subcarriers for a path can be represented as $\Delta\phi = 2\pi(f_0 + \Delta f \frac{d}{c}) + \lambda$, where d is the range, Δf is the subcarrier bandwidth, and λ is the random phase error due to packet boundary detection and sampling frequency offset. The phase error λ can be canceled out by computing the range difference between the LoS and NLoS paths as $d_{\text{NLoS}} - d_{\text{LoS}} = \frac{c}{2\pi\Delta f}(\Delta\phi_{\text{NLoS}} - \Delta\phi_{\text{LoS}})$, where $\Delta\phi_{\text{LoS}}$ and $\Delta\phi_{\text{NLoS}}$ can be estimated from the CSI.

Indoor Layout Representation Using SDF. Real-world indoor layouts are complex, featuring irregular layouts and diverse structural elements that are difficult to model. To address this, we adopt a signed distance function (SDF) to provide a continuous and differentiable representation of the layout. For any point in space, the SDF assigns it the signed distance to the nearest surface, and the set of points with an SDF value of zero forms a water-tight surface, which can be used to represent walls in indoor layouts (Fig. 4). We discretize the layout into voxels and maintain a discrete voxel-grid SDF. Compared with neural-represented SDFs, this voxel-based

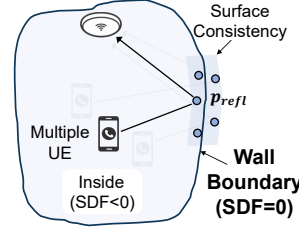


Figure 4: SDF example

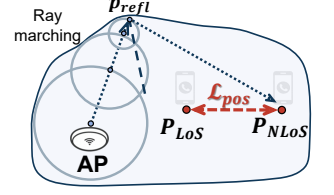


Figure 5: Position loss

formulation requires less data and is more robust to noisy reflection points.

SDF Training Loss. The key idea is to use LoS-derived UE positions to supervise reflections, driving reflection points toward correct locations. As illustrated in Fig. 5, given a UE, we have obtained its LoS-based position P_{LoS} from infrastructure mapping and can derive a new position P_{NLoS} by ray marching along its reflection path in the SDF-based layout. We optimize SDF to minimize the discrepancy between the two UE location estimates. Since P_{NLoS} depends on the wall locations and orientations, this optimization gradually refines the wall layout. Although individual reflection points may be noisy, the SDF fuses large numbers of such observations into a continuous and best-matching surface. We define the position loss as

$$\mathcal{L}_{\text{pos}} = \sum_{l \in \mathcal{U}} \|P_{\text{LoS}}^l - P_{\text{NLoS}}^l\|. \quad (4)$$

Moreover, to maintain a valid SDF, we add a regularization term that enforces a unit-norm gradient:

$$\mathcal{L}_{\text{sdf}} = \sum_{p \in \mathcal{V}} (\|\nabla \text{SDF}(p)\| - 1)^2, \quad (5)$$

where $\mathcal{V}$ denotes the set of voxel vertices.

Since no prior knowledge of layout is assumed, initializing SDF is challenging. To make WISEMap robust to arbitrary initialization, we introduce two loss terms to address SDF over-estimation and under-estimation. Overestimation is penalized when the distance between AP and reflection point exceeds total reflection path length:

$$\mathcal{L}_{\text{refl}} = \sum_{l \in \mathcal{P}} \max((L_{\text{tot}}^l - L_1^l) - \text{SDF}(p_{\text{refl}}^l), 0), \quad (6)$$

where $\mathcal{P}$ denotes the set of AP-UE pairs, L_{tot}^l and L_1^l denote the total reflection path length and the AP-to-reflection distance for the l -th AP-UE pair, respectively. Underestimation is penalized by enforcing that APs and UEs remain inside the reconstructed layout:

$$\mathcal{L}_{\text{out}} = \sum_{l \in \mathcal{UE}} \max(\text{SDF}(p_{\text{UE}}^l), 0). \quad (7)$$

The final objective is

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{pos}} + \lambda_2 \mathcal{L}_{\text{sdf}} + \lambda_3 \mathcal{L}_{\text{out}} + \lambda_4 \mathcal{L}_{\text{refl}}, \quad (8)$$

where λ_i weights the loss terms, and the contribution of each loss term is evaluated via an ablation study in §4.6.

Region Reconstruction. As mentioned earlier, for layout reconstruction, we partition the scene by AP coverage and let each AP reconstruct its local region. Specifically, this partitioning is completed by assigning each UE to the AP from which it receives

the strongest RSSI. Doing so improves the accuracy of AoA and range estimation of reflection path because estimation degrades significantly when signal strength is weak. By restricting each UE's reflection path estimation to the AP with the strongest RSSI, we ensure that reconstruction relies only on the most reliable reflection paths. Then the indoor layout can be reconstructed by minimizing the SDF training loss. However, it is worth noting that effective reconstruction within any region requires a sufficient number of reflection points to adequately constrain the walls. In our setting, APs are mounted on the ceiling, while UEs are located at typical handheld height. The NLoS paths reflecting off the ceiling or the floor are close to the LoS path and therefore difficult to resolve. Therefore, the reflection points of resolvable NLoS paths tend to lie around the midpoint between the AP and UE heights, denoted as h_{UE} and h_{AP} . As a result, the reconstruction is most accurate around the height $\frac{h_{UE}+h_{AP}}{2}$. We therefore use the SDF slice at this height as the reconstructed layout.

2.4 Region Fusion

To assemble the per-AP regions into a global layout, we first infer their spatial relationships, then fuse them accordingly, and finally regularize the fused regions.

Determining the Relationships Between Regions. We define two types of spatial relationships between regions: connected and disconnected. Two regions are considered *connected* if no walls separate them, and *disconnected* otherwise. In the first iteration, when the layout is not yet available, we assume that all regions are initially treated as connected. If two APs are in fact separated by walls, the LoS paths from their common UEs to one of the APs will be blocked. Such blockage reduces RSSI, causing larger UE-AP distance, and consequently overestimates inter-AP distance. As a result, when each AP reconstructs its local region in the first iteration, the overestimated inter-AP distance appears as a large spatial gap between the reconstructed regions. We use this gap as a reliable indicator that the two regions are disconnected; otherwise, they remain connected.

Fusing Regions. Let SDF_1 and SDF_2 denote the SDFs of two regions. If two regions are connected, we fuse them using the minimum of the two SDFs, i.e.,

$$SDF_{fused} = \min(SDF_1, SDF_2). \quad (9)$$

If two regions are disconnected while sharing an intersection area, we introduce a separating wall and fuse the regions as

$$SDF_{fused} = \begin{cases} \alpha/2 - |SDF_1 - SDF_2|, & \text{if } SDF_1 < 0 \text{ \& } SDF_2 < 0, \\ \min(SDF_1, SDF_2), & \text{otherwise,} \end{cases} \quad (10)$$

where α specifies the wall thickness and is set to a small positive value.

Region Regularization. Raw SDF boundaries are often irregular in shape due to measurement noise. Considering that indoor wall surfaces are typically flat, we simplify the geometric representation by fitting rectangles to the reconstructed layout. We define inside area as $A_{inside} = \{p \mid SDF(p) < 0\}$ and outside area as $A_{outside} = \{p \mid SDF(p) \geq 0\}$. A rectangle is parameterized as $\mathcal{R} = [x_{min}, x_{max}] \times [y_{min}, y_{max}]$. For each group of connected regions, we find a set of rectangles that maximize the coverage of

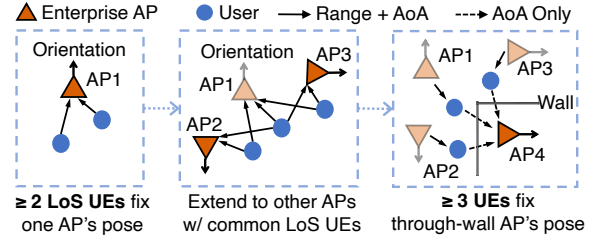


Figure 6: Identifiability of infrastructure and layout mapping (AP pose = AP location and orientation)

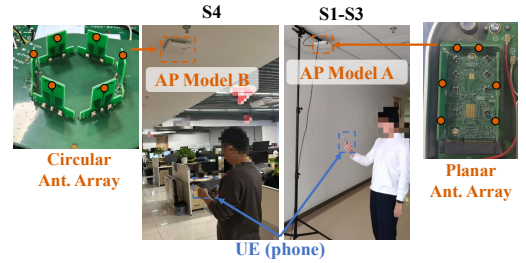


Figure 7: Antenna layouts and experiment setup

inside area while minimizing the inclusion of outside area. We begin by selecting the rectangle that minimizes the following objective: $\min_{\mathcal{R}} \mathcal{L} = -A_{inside} + A_{outside}$. After obtaining the first rectangle, we remove its covered portion of the SDF region and repeat the process to extract additional rectangles until the total covered area exceeds $\beta\%$ of the area of the connected regions. We merge all extracted rectangles to form the refined shape of the connected regions.

2.5 Convergence and Identifiability

Key observation. Wall-induced attenuation creates significant RSSI drops, enabling WISEMap to stably infer wall structures. The reconstructed wall layout is then used to distinguish LoS and through-wall paths, which allows proper use of paths for localization (i.e., range + AoA for LoS paths, AoA-only for NLoS paths, as in Fig. 3).

Convergence. As WISEMap alternates between optimizing infrastructure mapping and layout mapping, convergence hinges on the mutual influence between the two modules stabilizing. The influence of layout mapping on infrastructure mapping is mainly through path occlusions. Based on the above observation, walls introduce significant RSSI attenuation, which induces a clear separation of AP regions and enables their reliable reconstruction during layout mapping. Consequently, the path attributes (LoS or NLoS) in infrastructure mapping progressively stabilize, driving its convergence. The resulting infrastructure map further constrains layout mapping, facilitating the reconstruction of walls that contribute to signal reflections.

Identifiability. It can be proved that the joint problem of infrastructure and layout mapping is identifiable given a sufficient number of shared UE observations. As shown in Fig. 6, in principle, two LoS UEs suffice to determine an AP's pose (location and orientation). In practice, more UEs are needed to improve robustness against measurement noise. This approach extends naturally to

localizing other APs with common LoS UEs, thereby recovering the relative poses between APs. For APs separated by walls, at least three UEs are required for AoA-based localization. When AP poses are determined, the layout can be recovered by identifying wall occlusions and reflections, both of which can be uniquely identified by CSI and RSSI measurements. Consequently, the joint problem is identifiable.

3 Implementation

Framework. Our framework is primarily implemented in Python. We leverage PyTorch to accelerate matrix-intensive operations in RAP-MUSIC and build the region reconstruction module based on SDFDiff [11], a differentiable signed distance field framework. We voxelize the grid as a $128 \times 128 \times 128$ matrix and implement sphere tracing as a custom CUDA kernel, enabling massively parallel ray marching across voxels.

Hardware Platform. We select enterprise Wi-Fi devices for evaluation. Specifically, we use two enterprise Ruijie AP models: Model A and Model B¹. Both APs are equipped with six antennas: Model A adopts a planar array, while Model B uses a circular array (Fig. 7). Both models support IEEE 802.11ac and operate in the 5GHz band. We extract CSI using the vendor-provided tool. As user devices, we employ Huawei Mate 70 and Xiaomi 17 smartphones.

Height Configuration. Since the antenna arrays can estimate elevation angles, WISEMap estimates the AP height by combining elevation measurements from multiple UEs. The AP heights in S1–S4 are 2.9, 2.9, 3.2, and 3.2 m, respectively, and the average AP height estimation error is 0.813 m. We assume a UE height of 1.2 m, representing a typical handheld height. This fixed-height assumption may introduce additional localization errors when devices are carried at substantially different heights.

CSI Collection. Due to hardware constraints, both Ruijie APs can collect CSI only from the first antenna and one additional antenna at a time. Consequently, obtaining CSI from all six antennas requires five sequential measurements. Using the first antenna as the reference, we compute the inter-antenna phase differences for each measurement and then combine the results to construct a consistent multi-antenna CSI representation. We further compensate for hardware-induced antenna phase offsets using factory calibration parameters provided for the Ruijie APs. We observe that APs of the same model exhibit highly consistent phase calibration coefficients. Therefore, we use a shared set of calibration parameters for APs of the same model.

4 Evaluation

We evaluate WISEMap across four diverse deployment scenarios. Our evaluation covers three core tasks: layout reconstruction (§4.2), AP localization (§4.3), and UE localization (§4.4). We further analyze key impacting factors (§4.5) and examine critical design choices, including the SDF loss function (§4.6) and the iterative refinement process (§4.7).

4.1 Testbed Setups

We conduct experiments in four scenarios, with all APs mounted on the ceiling. In S1–S3, smartphones are handheld and remain

static, with around 20 CSI measurements collected at each location. In S4, smartphones are handheld by users while walking, with around 50 CSI measurements collected per second. For APs, S1–S3 use AP Model A, while S4 is an operational network that uses AP Model B. The environment layouts are shown in Fig. 8. These scenarios progressively evaluate WISEMap across indoor layouts of increasing complexity. The **loop corridor [S1]** features a $1.5 \text{ m} \times 6 \text{ m}$ load-bearing column at the center of a 90 m^2 area, with an AP density of $1/30 \text{ m}^2$ and a data density of 2.5 samples/m^2 . It evaluates WISEMap’s ability to reconstruct isolated, small-scale structures. The **meeting room [S2]** covers 120 m^2 (AP density of $1/40 \text{ m}^2$), contains two physically separated regions and has a data density of 1.8 samples/m^2 . This scenario evaluates WISEMap’s ability to separate disconnected regions. The **office floor [S3]** is a 540 m^2 enterprise office area with three separate rooms, with a data density of 0.2 samples/m^2 . Six APs are sparsely deployed (AP density of $1/90 \text{ m}^2$). This scenario evaluates WISEMap in a realistic enterprise environment at moderate scale. The **building floor [S4]** covers a 1800 m^2 enterprise floor with multiple rooms and corridors (AP density of $1/150 \text{ m}^2$). As users move in this scenario, the data collection lasts for 30 minutes. This scenario uses an operational enterprise Wi-Fi AP network to evaluate the scalability of WISEMap under large coverage and increased structural complexity.

4.2 Layout Reconstruction

We evaluate the core feature of WISEMap to reconstruct indoor layouts and compare it with the baselines.

Baselines. As existing reconstruction methods cannot map the layout without prior knowledge of AP and UE locations, our comparison focuses on their capabilities of layout mapping. For fair comparison, we present the performance of a variant of WISEMap, **WISEMap-GT**, that uses ground-truth AP and UE locations and orientations. We choose three state-of-the-art approaches as baselines:

- **NeRF²** [31] learns the propagation characteristics of electromagnetic wave with volumetric neural fields, allowing the environment to be inferred from the voxel densities.
- **Geometry-based methods** [10, 14] identify reflection points by exploring the geometric relations between APs, UEs, and then cluster and fit these reflection points to the best-matching reflecting surfaces.
- **EchoNeRF** [2] improves NeRF² by introducing an explicit wall-reflection model, leading to more accurate reconstruction of reflecting surfaces.

We present the reconstructed layouts in Fig. 9. Across all scenes, WISEMap successfully reconstructs the overall layouts, significantly outperforming the SOTA methods. Specifically, it identifies the load-bearing rectangular column at the corridor center in S1 and reconstructs separate areas and recovers their spatial relationships in S2–S4. In contrast, none of the baselines can correctly recover the overall layout. Although NeRF² excels at predicting spatial RSSI distributions, it fails to learn the underlying physical layout structure. EchoNeRF assumes precise RSSI measurements, which in practice fluctuate considerably. EchoNeRF recovers walls that obstruct LoS paths but fails to recover reflecting walls. Geometry-based methods depend heavily on accurate AoA and range estimation to infer

¹Due to commercial confidentiality, the exact model names are omitted.

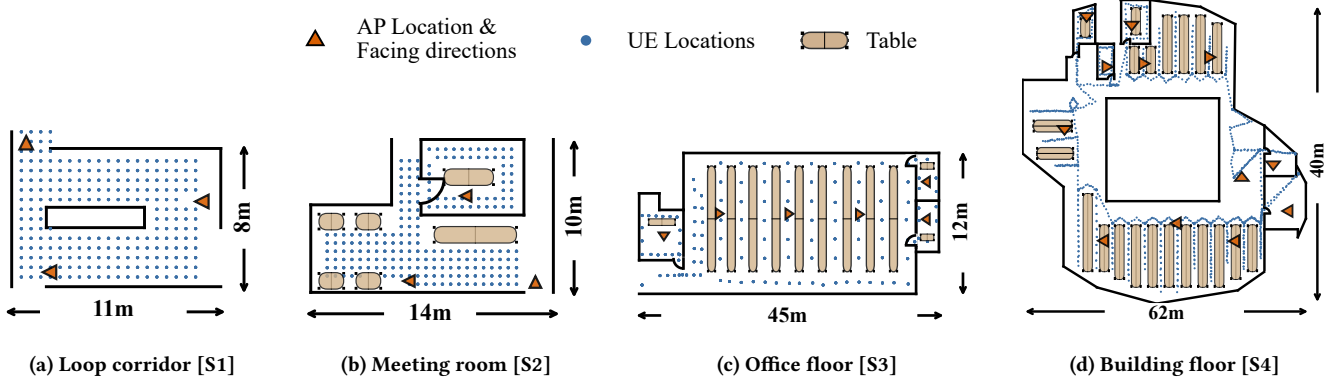


Figure 8: Four scenes used in experiments

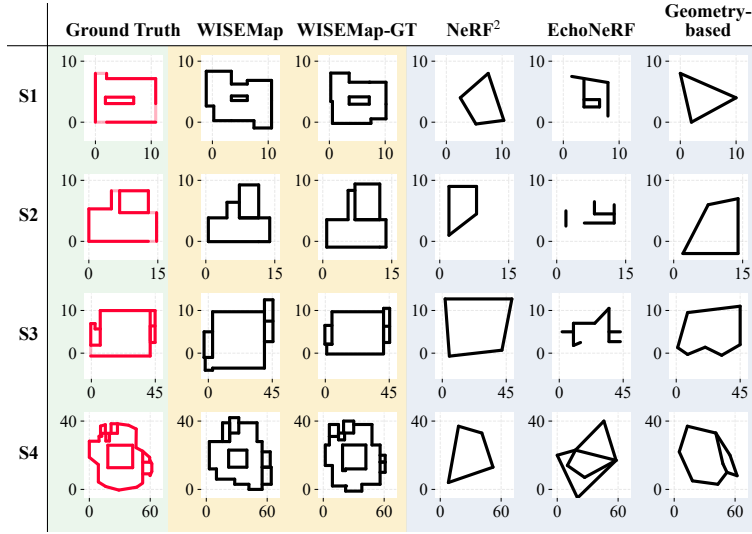


Figure 9: Layout reconstruction by different methods

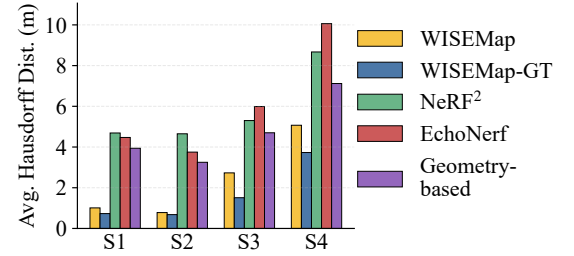


Figure 10: Layout error

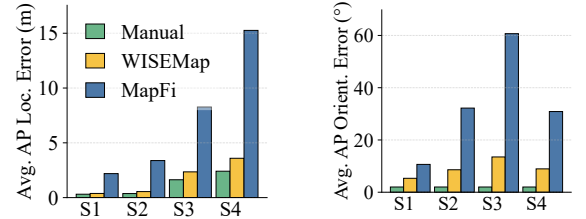


Figure 11: AP location and orientation error

reflection points. However, real-world Wi-Fi hardware introduces large errors in these estimates (more discussion in §4.5), causing the inferred reflection points to be matched with incorrect reflecting surfaces.

We further quantify the reconstruction accuracy using the Hausdorff distance between the reconstructed and the ground-truth layouts, which captures both global layout correctness and fine-grained boundary accuracy. As shown in Fig. 10, WISEMap-GT achieves the lowest Hausdorff distances among all methods, outperforming the baselines by 46–83%. Even when lacking the ground-truth AP and UE poses, WISEMap still reduces the reconstruction error by 31–77%. These results validate that WISEMap is effective at both the room-level layout and the wall-level geometry. The improvements arise from two aspects: (1) WISEMap asks each AP to reconstruct their nearby regions and leverage accurate AP localization to enforce correct region-level layout and (2) the SDF-based region reconstruction fits a flexible continuous surface that best

matches all reflection points observed at each AP, averaging out random errors in their locations. In the building floor [S4], WISEMap exhibits a noticeably larger Hausdorff distance of 5.1 m. This is primarily due to the scene’s large physical scale (62 m × 40 m) and complex office layout, which increases AP and UE localization errors and subsequently impacts reconstruction accuracy (more details in §4.3 and §4.4).

4.3 AP Self-localization

WISEMap enables APs to localize themselves within the layout, and these localization errors in turn influence reconstruction quality. We evaluate its AP localization accuracy against both a manual site survey and MapFi [26], the SOTA method that infers AP positions using inter-AP AoA estimates. For the manual site survey, we use the phone IMU to track the experimenter’s trajectory. The experimenter traverses all APs and records the estimated position when passing each AP, with the IMU periodically calibrated during the

survey. For AP orientation, the experimenter aligns the smartphone with the presumed reference direction of the AP enclosure and records its heading using the smartphone compass.

As shown in Fig. 11, WISEMap accurately estimates AP locations and orientations across all scenes. Compared with manual site surveys, WISEMap achieves comparable AP localization accuracy, while manual surveys generally achieve lower orientation errors due to the accurate orientation sensing of smartphones. However, this procedure measures the orientation inferred from the AP enclosure rather than directly sensing the orientation of the internal antenna array. Both evaluated AP models have symmetric enclosures, providing no reliable external cue for identifying the antenna array's reference direction. Therefore, when an AP is installed with an unexpected rotation relative to the presumed alignment, a manual survey may report an incorrect antenna orientation. Compared to MapFi, WISEMap reduces the location error by 67–88% and the orientation error by 57–73%. The average localization error in S4 is noticeably larger than in other scenes, primarily because S4 has a much lower AP density ($1/150 \text{ m}^2$). When APs are far apart in relatively large layouts, the weak received signal significantly degrades the inter-AP AoA estimation accuracy in MapFi. In contrast, WISEMap links neighboring APs through their common UEs, enabling AP localization over much larger separations and leveraging the collective observations from UEs. Moreover, with the reconstructed layout, WISEMap can differentiate direct LoS and through-wall paths and leverage their respective strengths accordingly. Comparing Fig. 11 and Fig. 10, it can be clearly seen that AP localization error is a major contributor to the layout reconstruction error. A more detailed analysis of AP localization error is provided in §4.5.

4.4 UE Localization

We evaluate WISEMap's UE localization accuracy and compare it with several popular localization methods to demonstrate its environment-aware localization capability.

Baselines. We choose the following methods as baselines.

- **SpotFi** [12] assumes the presence of a dominant LoS path to each UE and performs localization by combining AoA estimation with RSSI-based ranging.
- **UbiLocate** [23] measures the reliability of individual AP estimates on UE locations and determines the final UE location through a reliability-weighted centroid.
- **MapFi** [26] first estimates AP locations using inter-AP AoA estimates and then localizes UEs via triangulation.

UE Localization with True AP Poses. Since UbiLocate and SpotFi assume known AP poses (locations and orientations), we first evaluate WISEMap using ground-truth AP poses for fair comparison. As shown in Fig. 12a, WISEMap achieves significantly higher UE localization accuracy than the two baselines, reducing the average error by 13%–67% across all the scenes. This improvement is primarily because WISEMap identifies through-wall paths from SDF-based layout and discards their unreliable range estimates.

UE Localization without AP Pose Priors. To understand how the baselines perform in real-world scenarios where AP poses are unavailable, we evaluate UbiLocate and SpotFi using the estimated AP poses by WISEMap as ground truth, denoted as UbiLocate+ and

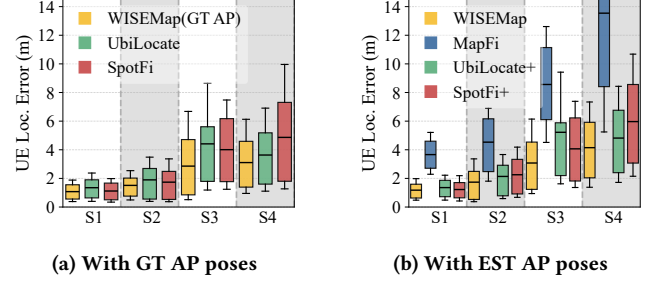


Figure 12: UE localization errors

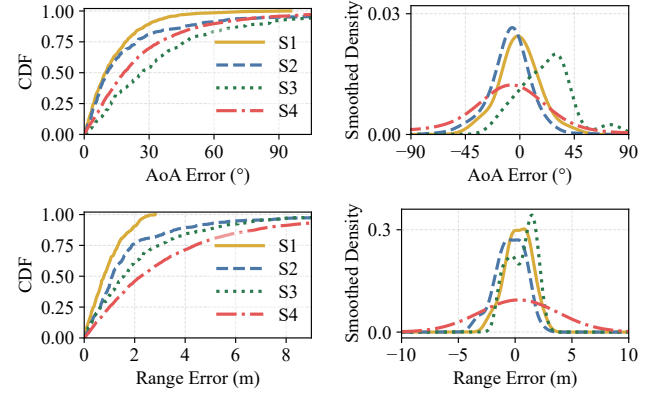


Figure 13: AoA and range error (CDF and Density)

SpotFi+. MapFi does not require prior AP information and uses its own AP pose estimates for comparison. WISEMap still significantly outperforms all baselines for most scenes, reducing UE localization error by 6%–81%.

4.5 Diving Deeper into Localization Errors

Since localization accuracy greatly depends on the AoA and range estimates derived from CSI and RSSI measurements, we analyze their estimation errors across the scenes.

As shown in the left part of Fig. 13, WISEMap exhibits noticeably larger AoA and ranging errors with long-tailed distributions in S3 and S4, due to the rich multipath and obstructions in real-world office environments. However, these errors are not systematically biased. Fig. 13 also shows the smoothed AoA and ranging error distributions for a random AP in each scene. The errors are approximately symmetric around zero, suggesting that they largely cancel out when aggregated across sufficient observations.

4.6 Ablation Study on SDF Loss

WISEMap's SDF optimization is guided by multiple loss terms that jointly enforce the geometric correctness of the reconstructed layout. To understand their individual contributions, we conduct an ablation study by removing each term in turn and measuring the resulting reconstruction errors. As shown in Fig. 14, excluding any loss term leads to noticeable degradation for the reconstructed meeting room [S2]. Quantitatively, removing any loss term will cause the average Hausdorff distance to increase (Fig. 14).

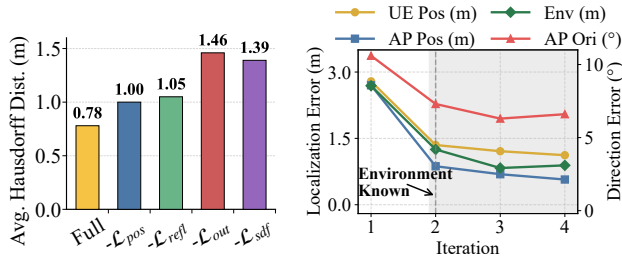


Figure 14: Hausdorff distance w/o certain loss **Figure 15: Iterative refinement process**

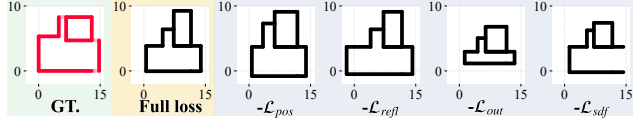


Figure 16: Reconstruction results for ablation study by removing one loss term

Specifically, as shown in Fig. 16, removing $\mathcal{L}_{pos}$ or $\mathcal{L}_{refl}$ causes the reconstructed walls to drift outward, overestimating the layout size. Removing $\mathcal{L}_{out}$ causes some APs and UEs not included within the reconstructed layout. Removing $\mathcal{L}_{sdf}$ degrades surface continuity and geometric consistency, resulting in fragmented, non-watertight surfaces and missing wall segments.

4.7 Progressive Refinement Process

WISEMap adopts an iterative alternating optimization framework, where early reconstructions provide coarse layout priors for AP and UE localization, and the resulting refined locations subsequently improve the later layout. As shown in Fig. 15, localization errors for both APs and UEs drop sharply after the first iteration once layout information is considered. Starting from iteration 2, the reconstructed layout enables WISEMap to identify through-wall paths and filter out RSSI measurements of through-wall paths. Fig. 17 visualizes this progressive process. In iteration 1, without knowledge of through-wall paths, WISEMap misplaces APs in the isolated meeting room due to wall blockages. After recognizing the walls in iteration 2, WISEMap identifies these NLoS paths and refines the AP positions, demonstrating the benefit of NLoS supervision for AP localization. The improved AP positions further reduce layout distortions. The results in iterations 3 and 4 remain nearly identical, indicating that both localization and reconstruction converge within 3 iterations. Finally, using region regularization, the final wall layout is obtained.

We further conduct an ablation study on infrastructure mapping to quantify the contribution of LoS and NLoS supervision in Eq. 3. We compare the complete WISEMap objective with two variants that remove the LoS loss and the NLoS loss, respectively, while keeping all other settings unchanged. With the complete objective, WISEMap achieves an average AP localization error of 0.57 m and an orientation error of 6.60°. Removing the LoS loss increases the errors to 3.72 m and 9.5°, while removing the NLoS loss increases them to 2.70 m and 10.6°, respectively. These results confirm that

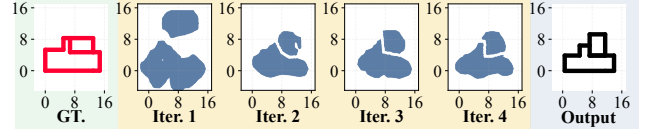


Figure 17: Progressive refinement process

both LoS and NLoS supervision provide important and complementary constraints for infrastructure mapping, and quantitatively demonstrate the contribution of NLoS supervision observed in the progressive refinement process.

5 Related Works

Automated Infrastructure Mapping. Most Wi-Fi systems simply assume that the infrastructure map is known, and only a few attempt to localize the APs. Methods such as MapFi [6, 26] estimate inter-AP AoA estimates, which are sensitive to blockage and multipath effects. To overcome this, LocAP [3] collects CSI measurements from numerous robot poses and achieves high AP localization accuracy, but requires dedicated robots and known trajectories. Recent enterprise solutions, such as Cisco AP Auto Locate and Aruba self-locating APs [7, 9], rely on GNSS and FTM ranging to determine AP positions, requiring specialized hardware support. RF-based SLAM systems have further explored estimation of infrastructure. CLAM [20] jointly estimates anchor locations and reflecting boundaries using mmWave measurements, while COMPAS [17] jointly estimates UE states and environmental features. However, these systems are designed for mmWave sensing and rely on dedicated measurement procedures or user-side state evolution. In contrast, WISEMap targets commodity enterprise Wi-Fi and passively aggregates noisy AP-side observations from regular user traffic. By fusing diverse UE observations across multiple enterprise APs, WISEMap simultaneously localizes the AP infrastructure and reconstructs the indoor layout without requiring specialized hardware, known user trajectories, or UE coordination.

Wi-Fi Localization. Early systems like SpotFi [12, 13] rely on LoS signals but degrade under blockage, as they do not explicitly model through-wall propagation. Recent approaches such as UbiLocate [23, 32] detect LoS blockage using channel statistics or cross-AP consistency. Learning-based methods [1, 4] extract richer features from CSI. However, these methods treat blockage independently for each AP-UE pair and lack a global view of the environment. In contrast, WISEMap explicitly reconstructs the indoor environment and leverages it to distinguish LoS and through-wall paths, enabling more consistent and accurate localization.

Layout Reconstruction. Indoor layout reconstruction is extensively studied with non-RF sensors. Visual methods leverage SDFs [5, 11, 21, 29, 33] for layout reconstruction. While non-RF methods achieve high reconstruction accuracy, they rely on specialized hardware and active measurements. WISEMap, in contrast, can leverage commodity APs already deployed and the regular Wi-Fi traffic between APs and UEs.

Previous studies have also explored using RF signals for layout reconstruction. Analytical RF models [10, 14] recover accurate wall surfaces but rely on accurate NLoS extraction, which seldom holds with commodity Wi-Fi due to noisy CSIs. To avoid explicit

NLoS extraction, neural RF models [2, 16, 31] learn an RF field, achieving high prediction accuracy of RSSIs but still failing to faithfully recover complete and accurate indoor layout. In contrast, WISEMap optimizes a SDF representation of the layout to find the best-matching indoor layout.

6 Limitations

Weak wall attenuation. WISEMap relies on sufficiently strong wall-induced attenuation as a useful signal for distinguishing through-wall paths from direct LoS paths. In our measurements, through-wall links exhibiting approximately 10 dB or larger additional attenuation can generally be identified, such as those caused by concrete walls and closed metal doors. However, its through-wall/LoS classification may become less reliable around open doors or glass windows, where the resulting signal attenuation characteristics can be similar to those of direct LoS paths.

Irregular wall layouts. WISEMap currently assumes that indoor layouts can be approximated by rectangular wall structures. While this assumption is applicable to many practical indoor environments, it may limit the reconstruction accuracy in environments that have highly irregular or curved wall geometries. Improving the robustness of path classification and extending the current layout representation to more general and flexible geometries are important directions for future work.

7 Conclusion

This paper presents WISEMap, an alternating optimization framework that achieves simultaneous mapping of Wi-Fi infrastructure and indoor layout using enterprise APs. By leveraging the interdependence between the two tasks, WISEMap can alternately improve each task using the refined output of the other. Our evaluations show that infrastructure mapping and layout reconstruction are strongly coupled, and that joint optimization of the two tasks significantly outperforms the baselines designed for each task separately. Looking ahead, we will also enhance WISEMap to handle crowdsourced data and move toward deployment at scale in enterprise Wi-Fi networks.

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